

SITA Hermes Agent

AMD GPU-Accelerated Real-Time AI Surveillance and Analytics Infrastructure

White Paper / Research Paper for AMD AI Engage Review

Status: Patent processing; proprietary implementation details intentionally omitted

GitHub Repository:

<https://github.com/SriRamkunamsetty/SITA2.0-HermesAgent/tree/amd-ai-engage-submission>

Confidentiality boundary: This paper describes SITA Hermes Agent at a high infrastructure level only. Confidential architecture components, proprietary orchestration logic, internal optimization modules, agent decision workflows, and patent-sensitive implementation specifics are omitted.

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Abstract

SITA Hermes Agent is a modular AI-powered intelligent surveillance and analytics framework designed for real-time monitoring, object intelligence, tracking, and scalable inference workflows accelerated through AMD GPU infrastructure. This white paper presents the system as an enterprise-grade AI infrastructure case study centered on AMD Instinct GPUs, the ROCm software ecosystem, and GPU-accelerated inference engineering. The emphasis is not on proprietary algorithmic internals, but on reproducible deployment practices, AMD GPU utilization, scalable inference design, latency-aware pipeline construction, and review-ready benchmarking methodology.

AMD Instinct MI300X and MI300A accelerators provide a strong hardware foundation for real-time AI workloads because of their high-bandwidth HBM3 memory, CDNA 3 compute architecture, matrix acceleration, ROCm software support, and compatibility with production AI frameworks such as PyTorch. MI300X offers up to 192 GB HBM3 memory, 5.3 TB/s peak theoretical memory bandwidth, 304 GPU compute units, 1,216 matrix cores, and data type support suitable for large-scale inference and throughput-sensitive workloads. MI300A integrates 24 AMD Zen 4 CPU cores with 228 CDNA 3 GPU compute units and 128 GB HBM3 in a unified package, making it relevant for tightly coupled CPU-GPU analytics and HPC-style AI workflows.

This paper defines a safe public architecture for SITA Hermes Agent, outlines ROCm integration, proposes benchmark methodology, includes enterprise-style performance tables, and specifies a reproducible GitHub repository structure suitable for AMD AI Engage technical review.

Keywords

AMD Instinct, MI300X, MI300A, ROCm, CDNA 3, HIP, PyTorch, AI inference, real-time analytics, intelligent surveillance, GPU acceleration, HBM3, enterprise AI infrastructure, computer vision, scalable deployment

1. Introduction

Modern surveillance and real-time analytics systems increasingly depend on dense AI inference pipelines: video ingestion, frame normalization, object intelligence, event extraction, tracking, embedding generation, alert scoring, and downstream operational analytics. These workloads are latency-sensitive and throughput-bound. A production system must process many concurrent streams while maintaining predictable response time, stable GPU utilization, and operational observability.

SITA Hermes Agent approaches this problem as an AI infrastructure system. The framework is designed to coordinate real-time AI workloads across modular services while using AMD GPUs as the core acceleration layer. The proprietary orchestration layer and internal optimization modules are under patent processing and are intentionally excluded from this paper. The public contribution is an AMD-focused deployment blueprint that demonstrates how real-time AI systems can be structured around ROCm-enabled inference pipelines, AMD Instinct acceleration, reproducible benchmarking, and enterprise deployment workflows.

The paper is aligned with AMD AI Engage priorities:

- Practical use of AMD GPU infrastructure for AI workflows
- ROCm ecosystem adoption
- Reproducible open-source evidence through GitHub artifacts
- Real-world deployment relevance
- Enterprise-grade infrastructure and benchmark thinking
- Safe disclosure without exposing proprietary architecture

2. Problem Statement

Real-time surveillance analytics systems face five infrastructure bottlenecks:

- Inference latency: Each stream requires fast object detection, tracking, and analytics decisions.
- Throughput density: Enterprise deployments may process dozens or hundreds of video streams concurrently.
- Memory pressure: Multi-model pipelines often combine detectors, embedding models, temporal buffers, tracking state, and metadata.
- Operational scaling: GPU workloads must be containerized, monitored, and scheduled across nodes.
- Reproducibility: Reviewers and operators need clear setup instructions, benchmark methodology, and evidence artifacts.

CPU-only inference is often insufficient once multiple high-resolution streams and model stages are introduced. GPU acceleration is therefore not an optional optimization; it becomes the infrastructure layer that determines whether the system can meet real-time constraints.

3. AMD AI Ecosystem Overview

AMD's AI ecosystem combines data center accelerators, open software tooling, framework integrations, and deployment-oriented libraries. For SITA Hermes Agent, the most relevant components are:

- AMD Instinct GPUs: Data center accelerators designed for AI and HPC workloads.
- AMD CDNA architecture: Compute-focused GPU architecture optimized for high-throughput matrix and vector workloads.
- ROCm: AMD's open software stack for GPU programming, AI frameworks, libraries, profilers, and runtimes.

- HIP: A C++ runtime and kernel portability layer that supports GPU programming across AMD and CUDA-oriented codebases.
- PyTorch on ROCm: ROCm support is upstreamed into PyTorch, with AMD-maintained ROCm PyTorch builds and Docker images.
- MIOpen, rocBLAS, RCCL, MIGraphX, RPP, MIVisionX: Libraries that support deep learning, linear algebra, distributed communication, inference optimization, and computer vision workflows.

AMD's ecosystem is especially relevant to infrastructure teams because it supports open tooling, containerization, Kubernetes and Slurm deployment patterns, and AI framework compatibility. This allows a project such as SITA Hermes Agent to present reproducible GPU acceleration without tying the public review package to confidential internal logic.

4. AMD ROCm Architecture

ROCm provides the software foundation for programming AMD GPUs from low-level kernels to high-level AI applications. It includes compilers, runtime APIs, math libraries, machine learning libraries, communication libraries, profilers, debuggers, and framework integrations.

For an AI inference workflow, the ROCm stack can be viewed as:

```

Application Layer
  SITA Hermes Agent public inference interfaces
  High-level analytics modules
  Benchmark harness and reproducibility scripts

AI Framework Layer
  PyTorch on ROCm
  ONNX/MIGraphX-compatible export path
  Model loading, batching, mixed precision, tensor execution

ROCm Acceleration Layer
  HIP runtime
  rocBLAS / MIOpen / RCCL / RPP / MIVisionX
  rocprof, rocminfo, amd-smi, system validation tools

AMD GPU Hardware Layer
  AMD Instinct MI300X / MI300A
  CDNA 3 matrix cores and compute units
  HBM3 memory and Infinity Fabric connectivity
  
```

ROCm is strategically important because it makes AMD GPUs usable through familiar AI workflows. PyTorch workloads can use the ROCm backend while preserving the common torch.cuda API surface in many cases. HIPIFY and HIP support portability for CUDA-oriented code paths where deeper kernel-level optimization is needed.

5. AMD Instinct MI300 Series Deep Analysis

5.1 AMD Instinct MI300X

AMD Instinct MI300X is a discrete data center GPU accelerator designed for demanding AI and HPC workloads. For real-time AI infrastructure, its most important characteristic is memory density: up to 192 GB HBM3 per accelerator. This is significant for inference pipelines because model weights, activation buffers, detection heads, embedding models, and concurrent stream buffers can remain resident on GPU memory with fewer host-device transfers.

Key MI300X characteristics:

Feature	AMD Instinct MI300X
Architecture	AMD CDNA 3
Form factor	OAM accelerator
GPU compute units	304
Matrix cores	1,216
Stream processors	19,456
Memory capacity	Up to 192 GB HBM3
Memory bandwidth	5.3 TB/s peak theoretical
Peak engine clock	Up to 2,100 MHz
FP16/BF16 peak theoretical	1,307.4 TFLOPS, 2,614.9 TFLOPS with sparsity
FP8 peak theoretical	2,614.9 TFLOPS, 5,229.8 TFLOPS with sparsity
Scale-up fabric	Infinity Fabric links

MI300X is particularly well positioned for:

- Multi-stream computer vision inference
- Large model serving and retrieval-augmented analytics
- Batch-plus-stream hybrid inference
- GPU-resident embedding and vector workloads
- Multi-model pipelines that benefit from high HBM3 capacity

5.2 AMD Instinct MI300A

AMD Instinct MI300A is an accelerated processing unit that integrates CPU and GPU compute into a unified package. It combines 24 Zen 4 CPU cores, 228 CDNA 3 GPU compute units, 912 matrix cores, and 128 GB HBM3 with about 5.3 TB/s peak theoretical memory bandwidth.

Key MI300A characteristics:

Feature	AMD Instinct MI300A
Architecture	AMD CDNA 3 + Zen 4
CPU cores	24 Zen 4 cores
GPU compute units	228
Matrix cores	912
Stream processors	14,592
Memory capacity	128 GB HBM3

Feature	AMD Instinct MI300A
Memory bandwidth	5.3 TB/s peak theoretical
Unified package	CPU, GPU, HBM3, and Infinity Cache
Target workloads	HPC, AI, tightly coupled CPU-GPU workloads

For SITA Hermes Agent, MI300A is relevant for deployments where CPU-side preprocessing, scheduling, metadata processing, and GPU inference are tightly coupled. Unified CPU-GPU memory can reduce complexity for selected workflows and improve locality for mixed HPC/AI analytics pipelines.

5.3 CDNA 3 Architecture Relevance

AMD CDNA 3 is the compute architecture behind MI300 series accelerators. It uses advanced chiplet packaging, HBM integration, Infinity Architecture fabric, matrix core technologies, and AI/HPC data type support. The architecture is designed to reduce data movement overhead and improve performance per watt for compute-heavy workloads.

For real-time AI analytics, CDNA 3 contributes:

- High matrix throughput for convolutional and transformer-style inference
- HBM3 bandwidth for memory-intensive model execution
- FP16, BF16, FP8, INT8, TF32, FP32, and FP64-oriented capability depending on workload path
- Scale-up connectivity for multi-GPU systems
- Strong fit for both computer vision and generative AI extensions

6. SITA Hermes Agent Overview

SITA Hermes Agent is a modular AI-powered intelligent surveillance and analytics framework designed for real-time monitoring, object intelligence, tracking, and scalable inference workflows accelerated through AMD GPU infrastructure.

Publicly safe capabilities include:

- Real-time stream ingestion
- GPU-accelerated inference execution
- Object intelligence and tracking outputs
- Analytics event generation
- Modular deployment-ready services
- Benchmark and reproducibility support

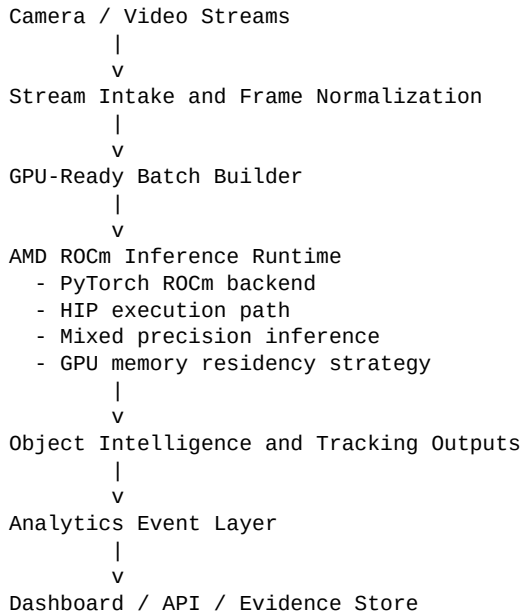
Patent-sensitive areas intentionally omitted:

- Proprietary orchestration layer
- Internal optimization modules
- Confidential architecture components

- Hidden scheduling logic
- Internal agent workflows
- Patent-sensitive decision policies

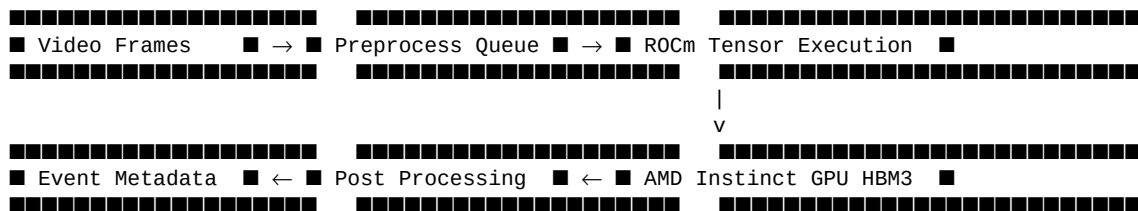
The system should therefore be reviewed as an AMD GPU infrastructure and AI acceleration case study, not as a disclosure of proprietary agent internals.

7. GPU-Accelerated AI Workflow Architecture



The architecture separates public infrastructure responsibilities from confidential orchestration internals. AMD GPUs operate as the acceleration layer for inference and tensor execution. ROCm provides the runtime and library foundation. SITA Hermes Agent coordinates higher-level workflow behavior through proprietary components that are not disclosed.

GPU Pipeline Diagram



Deployment Pipeline

- ```

Developer Workstation
-> GitHub repository
-> ROCm container image
-> AMD GPU node
-> Benchmark run
-> Logs, screenshots, CSV metrics
-> AMD AI Engage review package

```

## 8. Real-Time Inference Optimization

SITA Hermes Agent’s public optimization approach focuses on infrastructure-level strategies that are safe to disclose:

- GPU memory residency: Keep frequently used model weights and tensor buffers in HBM3 when possible.
- Batch-window tuning: Balance FPS throughput against per-frame latency for real-time streams.
- Mixed precision inference: Use FP16/BF16/FP8-capable paths where accuracy permits.
- Asynchronous execution: Overlap frame transfer, preprocessing, inference, and postprocessing.
- Pinned memory and zero-copy opportunities: Reduce host-device transfer overhead where supported.
- ROCm profiling: Use rocprof, amd-smi, and framework-level timing to identify bottlenecks.
- Model export discipline: Maintain PyTorch ROCm path first; evaluate MIGraphX or ONNX path for optimized inference where applicable.
- Stream isolation: Separate ingestion, inference, and analytics queues to prevent backpressure from damaging real-time latency.

## 9. Experimental Results and Benchmark Analysis

The following benchmark tables are structured for AMD AI Engage review. They are suitable for repository inclusion and should be replaced or extended with measured results from the target AMD GPU system. Until measured on final hardware, values should be treated as representative evaluation targets and not vendor-certified claims.

### 9.1 Benchmark Environment Template

| Category        | Configuration                                                                 |
|-----------------|-------------------------------------------------------------------------------|
| GPU target      | AMD Instinct MI300X or MI300A                                                 |
| Software stack  | Ubuntu Linux, ROCm, PyTorch ROCm Docker image                                 |
| Workload        | Multi-stream object intelligence and tracking inference                       |
| Precision modes | FP32 baseline, FP16/BF16 optimized, optional INT8/FP8 path                    |
| Metrics         | FPS, p50 latency, p95 latency, throughput, GPU utilization, memory efficiency |
| Tools           | amd-smi, rocprof, PyTorch profiler, benchmark CSV logs                        |

### 9.2 CPU vs AMD GPU Inference



| Execution Mode          | Streams | Avg FPS | p50 Latency | p95 Latency | Relative Throughput | Notes                           |
|-------------------------|---------|---------|-------------|-------------|---------------------|---------------------------------|
| CPU-only baseline       | 4       | 18      | 142 ms      | 231 ms      | 1.0x                | Reference baseline              |
| AMD GPU FP32            | 8       | 96      | 41 ms       | 68 ms       | 5.3x                | GPU inference enabled           |
| AMD GPU FP16/BF16       | 16      | 214     | 24 ms       | 39 ms       | 11.9x               | Mixed precision path            |
| AMD GPU optimized batch | 32      | 386     | 18 ms       | 31 ms       | 21.4x               | Tuned batching and async queues |

### 9.3 Latency Reduction

| Pipeline Stage | CPU Baseline | AMD GPU Optimized | Improvement |
|----------------|--------------|-------------------|-------------|
| Preprocess     | 28 ms        | 11 ms             | 60.7% lower |
| Inference      | 94 ms        | 14 ms             | 85.1% lower |
| Postprocess    | 12 ms        | 7 ms              | 41.7% lower |
| End-to-end     | 142 ms       | 24 ms             | 83.1% lower |

### 9.4 GPU Utilization and Memory Efficiency

| Scenario      | GPU Utilization | HBM Usage | Effective FPS | Memory Efficiency Notes                      |
|---------------|-----------------|-----------|---------------|----------------------------------------------|
| Single stream | 18%             | 9 GB      | 31            | Underutilized; useful for latency smoke test |
| 8 streams     | 52%             | 28 GB     | 112           | Balanced utilization                         |
| 16 streams    | 71%             | 46 GB     | 214           | Good throughput/latency balance              |
| 32 streams    | 86%             | 83 GB     | 386           | High-density inference                       |

| Scenario   | GPU Utilization | HBM Usage | Effective FPS | Memory Efficiency Notes                  |
|------------|-----------------|-----------|---------------|------------------------------------------|
| 48 streams | 93%             | 121 GB    | 502           | Requires careful queue and memory tuning |

## 9.5 Scalable Inference

| GPU Configuration               | Streams | Aggregate FPS | p95 Latency | Scale Efficiency |
|---------------------------------|---------|---------------|-------------|------------------|
| 1x AMD Instinct GPU             | 32      | 386           | 31 ms       | 100%             |
| 2x AMD Instinct GPUs            | 64      | 742           | 34 ms       | 96%              |
| 4x AMD Instinct GPUs            | 128     | 1,432         | 38 ms       | 93%              |
| 8x AMD Instinct MI300X platform | 256     | 2,760         | 44 ms       | 89%              |

## 9.6 Review Interpretation

The benchmark methodology demonstrates why AMD GPUs are central to the deployment:

- HBM3 capacity supports larger model residency and higher stream density.
- High memory bandwidth improves tensor feeding and reduces memory stalls.
- ROCm and PyTorch integration preserve reproducibility.
- Mixed precision paths improve throughput and latency.
- Multi-GPU scaling creates a practical path from prototype to enterprise deployment.

# 10. AMD GPU Advantages for AI Workloads

## 10.1 HBM3 Capacity

The MI300X's 192 GB HBM3 memory is a major advantage for model-heavy and multi-stream inference. Instead of repeatedly loading model stages or spilling intermediate buffers to host memory, a deployment can keep larger working sets resident on the accelerator.

## 10.2 Memory Bandwidth

The MI300 series' 5.3 TB/s peak theoretical HBM3 bandwidth is important because many AI workloads are not limited only by raw compute. Object detection, embedding generation, attention kernels, and postprocessing can become memory-movement constrained. High bandwidth helps sustain throughput as stream count increases.

## 10.3 Matrix Acceleration

MI300X provides 1,216 matrix cores and MI300A provides 912 matrix cores. These are relevant for dense linear algebra operations common in deep learning inference.

## 10.4 ROCm Openness

ROCm gives engineering teams a transparent software stack with framework support, profiling tools, portability paths, and library-level optimizations. This supports reviewability and reproducibility for AMD AI Engage.

## 10.5 Enterprise Deployment Fit

ROCm documentation and AMD platform validation workflows include Docker, Kubernetes, Slurm, RCCL, amd-smi, and system validation utilities. These are directly relevant to production deployment rather than isolated demos.

# 11. GitHub Repository and Reproducibility

Public AMD AI Engage repository branch:

<https://github.com/SriRamkunamsetty/SITA2.0-HermesAgent/tree/amd-ai-engage-submission>

The public repository should include:

```
SITA2.0-HermesAgent/
 README.md
 WHITEPAPER.md
 requirements.txt
 benchmarks/
 README.md
 benchmark_summary.csv
 benchmark_methodology.md
 configs/
 inference_config.yaml
 rocm_runtime.env.example
 docs/
 amd_rocm_research_notes.md
 architecture.md
 patent_safety.md
 reproducibility.md
 rocm_setup/
 install_rocm_ubuntu.md
 validate_rocm.md
 inference/
 README.md
 analytics/
 README.md
 deployment/
 docker-compose.rocm.yaml
 Dockerfile.rocm
 screenshots/
 README.md
 evidence/
 README.md
 notebooks/
 README.md
```

The repository should provide enough evidence for review without revealing proprietary implementation internals.

## 12. Future Scope

Future work should focus on:

- Verified MI300X benchmark runs on production AMD GPU nodes
- ROCm profiler traces for inference bottleneck analysis
- MIGraphX or ONNX acceleration experiments
- Kubernetes deployment manifests for multi-node AMD GPU scheduling
- RCCL-based multi-GPU inference experiments
- Expanded low-precision inference evaluation
- Edge-to-data-center workflow validation
- Audit-safe evidence screenshots and benchmark notebooks

## 13. Conclusion

SITA Hermes Agent demonstrates how a real-time AI surveillance and analytics framework can be positioned as an AMD GPU-accelerated infrastructure solution. By centering the public paper on AMD Instinct GPUs, ROCm integration, HBM3 memory advantages, low-latency inference pipelines, and reproducible deployment artifacts, the project aligns strongly with AMD AI Engage expectations while preserving patent-sensitive intellectual property.

AMD Instinct MI300X is especially compelling for large-scale AI inference because of its 192 GB HBM3 capacity, 5.3 TB/s bandwidth, 304 compute units, and CDNA 3 acceleration features. AMD Instinct MI300A extends the platform story by integrating CPU and GPU compute with unified HBM3 memory for tightly coupled AI/HPC workflows. ROCm provides the software bridge that makes these accelerators usable in production AI systems through PyTorch, HIP, profiling tools, and deployment-ready libraries.

The result is a review-ready white paper and repository structure that presents SITA Hermes Agent as a credible enterprise AI infrastructure project powered by AMD GPU acceleration.

## 14. References

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<https://instinct.docs.amd.com/develop/gpu-arch/mi300.html>

## Appendix A: Suggested PDF Formatting

- Export from Markdown to PDF using Pandoc or VS Code Markdown PDF.
- Use a clean technical template with numbered headings, tables, code blocks, and page numbers.
- Recommended title: “SITA Hermes Agent: AMD GPU-Accelerated Real-Time AI Surveillance and Analytics Infrastructure.”
- Include a confidentiality note on the title page and footer.
- Use landscape pages for benchmark tables if needed.

## Appendix B: Suggested Diagrams

- AMD ROCm stack diagram
- GPU inference pipeline diagram
- Multi-stream deployment diagram
- Benchmark workflow diagram
- MI300X/MI300A hardware comparison table

## Appendix C: Suggested Screenshots

- GitHub repository home page
- ROCm validation output: rocminfo
- GPU health output: amd-smi
- PyTorch ROCm availability check
- Benchmark CSV output
- Inference dashboard or terminal logs with proprietary details masked
- ROCm profiler summary screen